Experiment 2: β-galactosidase kinetics

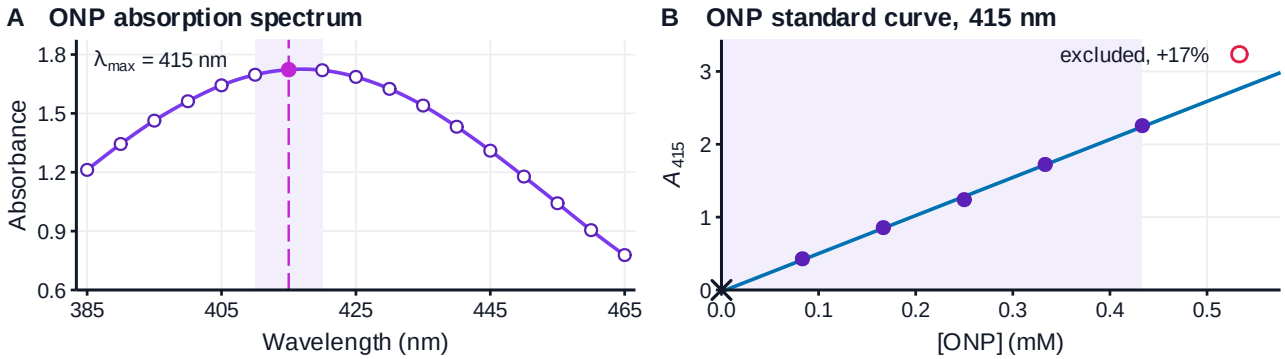
Figures (main and supplementary) at publication quality

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Main figures are drawn from the reference dataset (course-provided measurements used for the Results section at the instructor's direction); supplementary figures hold the experimental dataset (our own printouts), diagnostics and sensitivity analyses. Conventions: filled symbol, observation used for V_0; open symbol, observation retained but not used; solid line, fitted initial window; dashed segment from the origin, origin-assumed 0–2 min average rate (A_{415}(2 min)/2 min with A_{415} = 0 assumed at t = 0, where no reading exists); dotted line, extrapolation; † marks such an estimate, an average over the first interval rather than a measured instantaneous initial velocity. Each condition was measured once, so no error bars are shown. Full legends are in the course report and in the journal-style version; the vector PDF of every figure is in `figures/`.

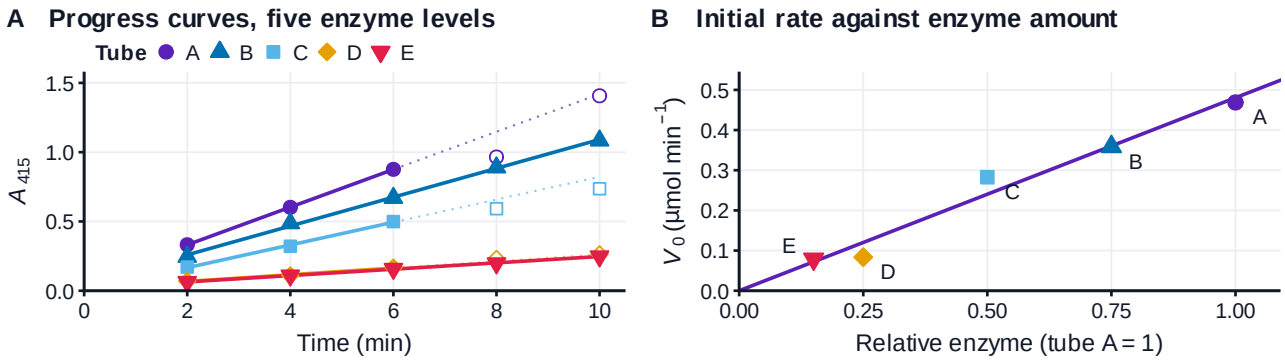
Contents: Figure 1 (ONP spectrum and standard curve), Figure 2 (enzyme dilution), Figure 3 (progress curves without and with IPTG), Figure 4 (saturation kinetics, IPTG inhibition and Lineweaver–Burk plots), graphical abstract, Figures S1–S4, reference diagram for Question 1.

Figure 1. ONP absorbs maximally at 415 nm and obeys the Beer–Lambert law to 0.43 mM



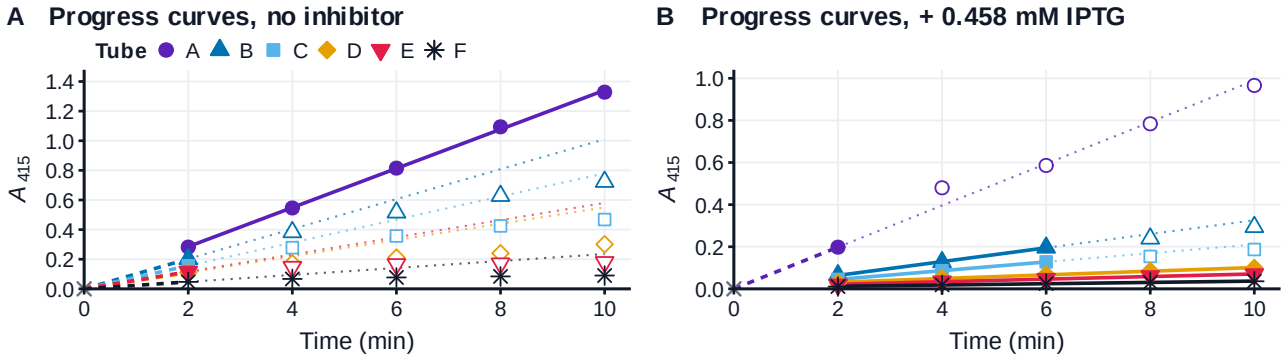
(A) Absorbance spectrum of 0.333 mM ONP in the alkaline assay matrix; dashed line, maximum at 415 nm; shaded band, within 2% of the maximum. (B) Standard curve at 415 nm; filled points and line, least-squares calibration over the shaded range, $A_{415} = 5.222[\text{ONP}] - 0.022$, $R^2 = 0.9988$, $\varepsilon = 5.22 \pm 0.10$ mM^{-1} cm^{-1}; ×, zero-concentration point (not fitted); open point, excluded 0.533 mM standard (17% above the line). Reference dataset, Part I.

Figure 2. Initial velocity rose approximately in proportion to the amount of β-galactosidase



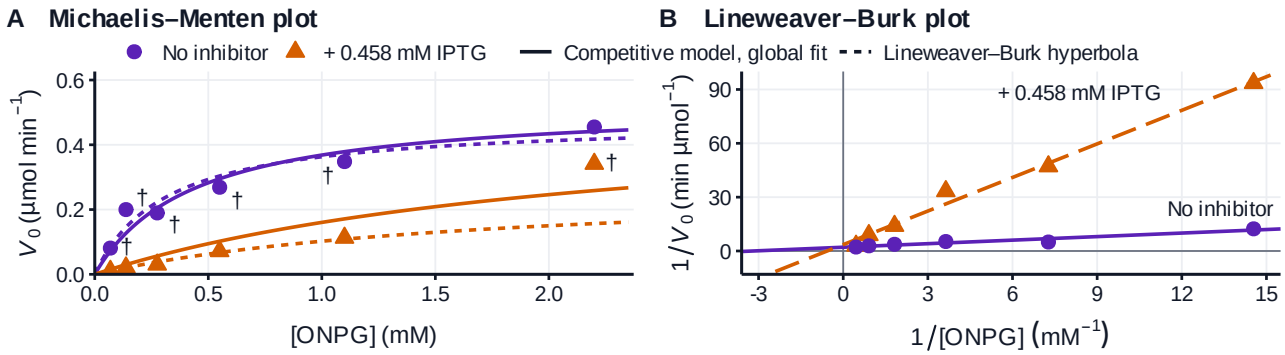
(A) Progress curves for 2.20 mM ONPG with 1.00 to 0.15 relative enzyme (tubes A–E); lines, least-squares fits to the initial linear window; filled points used, open points excluded, dotted extrapolation. (B) V_0 in µmol ONP min^{-1} per 12.0 mL assay against relative enzyme; solid line through the origin, slope 0.481, R^2 = 0.972; dotted, free-intercept fit (intercept -0.0006, p = 0.98). Relative rates 1.00, 0.77, 0.60, 0.18, 0.17 for relative enzyme 1.00, 0.75, 0.50, 0.25, 0.15. Reference dataset, Part II.

Figure 3. Product formation slowed as substrate fell and when IPTG was present



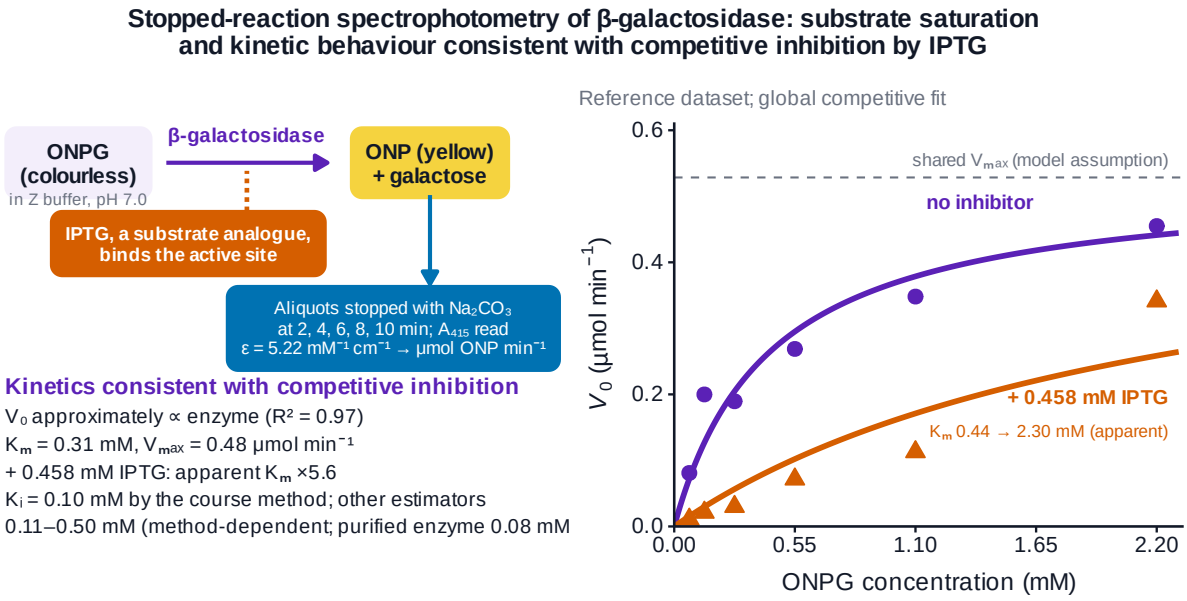
(A) Progress curves for 0.2 mL enzyme with 2.20 to 0.069 mM ONPG (tubes A–F, no inhibitor); tube A linear (regression); in B–F the later increments fell below the first, so V_0 is the origin-assumed 0–2 min average rate (dashed; ×, assumed origin, no reading at t = 0); open points not used; dotted, extrapolation. (B) The same six tubes with 0.458 mM IPTG on a smaller absorbance scale; tube A origin-assumed, tubes B–F by regression over the filled points. V_0 values in Table 4 of the course report. Reference dataset, Part III.

Figure 4. β-Galactosidase approached saturation with ONPG, and IPTG raised the apparent K_m



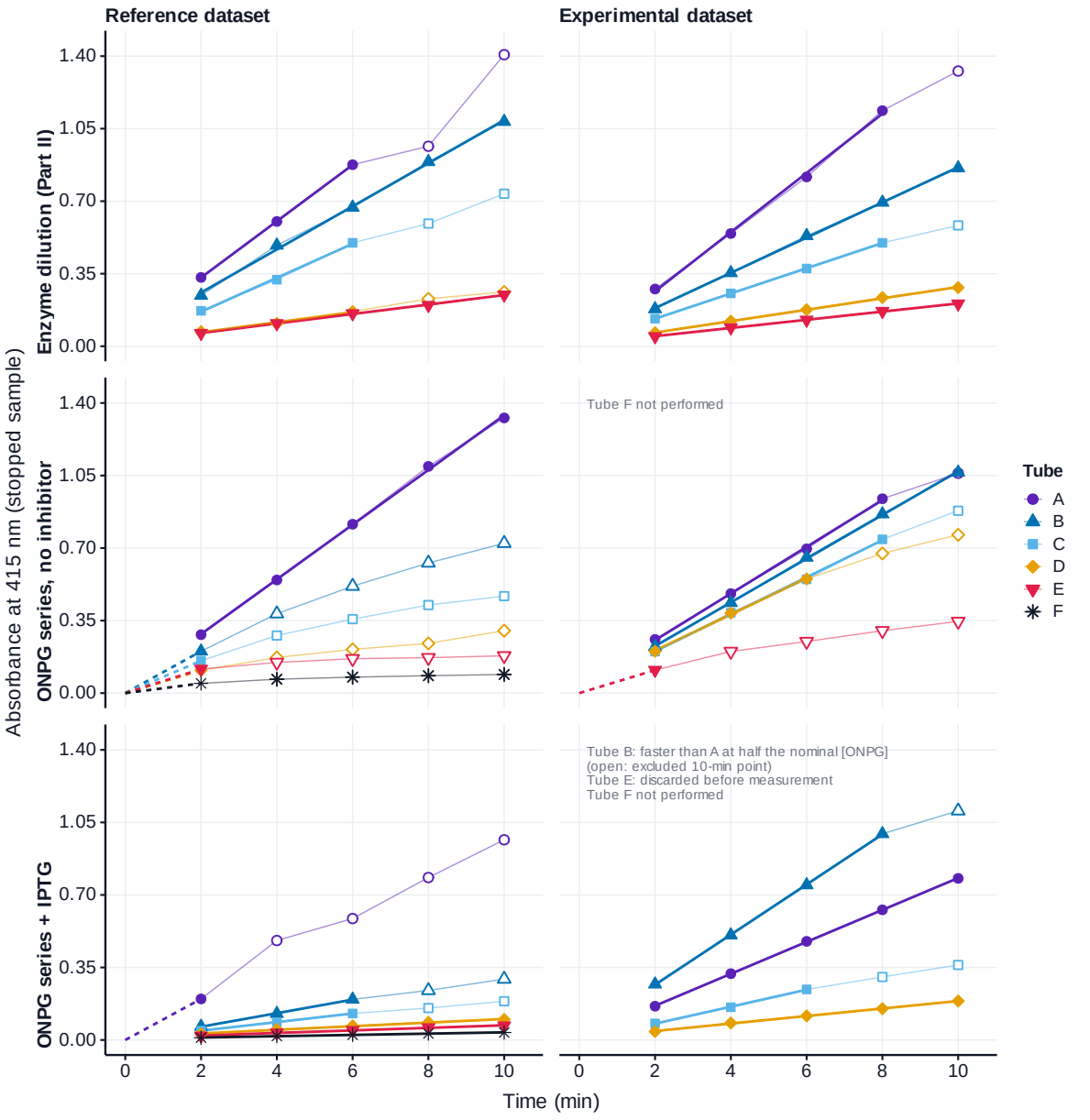
(A) V_0 against [ONPG] without inhibitor (circles) and with 0.458 mM IPTG (triangles), from the curves in Figure 3; dashed, hyperbolae from the Lineweaver–Burk lines; solid, one competitive model fitted to both series (shared V_{max} = 0.53 µmol min^{-1}, K_m = 0.44 mM, K_i = 0.107 ± 0.042 mM; the shared V_{max} is a model assumption); †, origin-assumed estimate. (B) Lineweaver–Burk plot: no inhibitor $1/V_0$ = 0.66(1/[S]) + 2.10 (K_m 0.31 mM, V_{max} 0.477); + IPTG $1/V_0$ = 6.24(1/[S]) + 3.55 ($K_{m,app}$ 1.76 mM, $V_{max,app}$ 0.282); K_i = 0.100 mM by the competitive relation. Reference dataset, Part III.

Graphical abstract



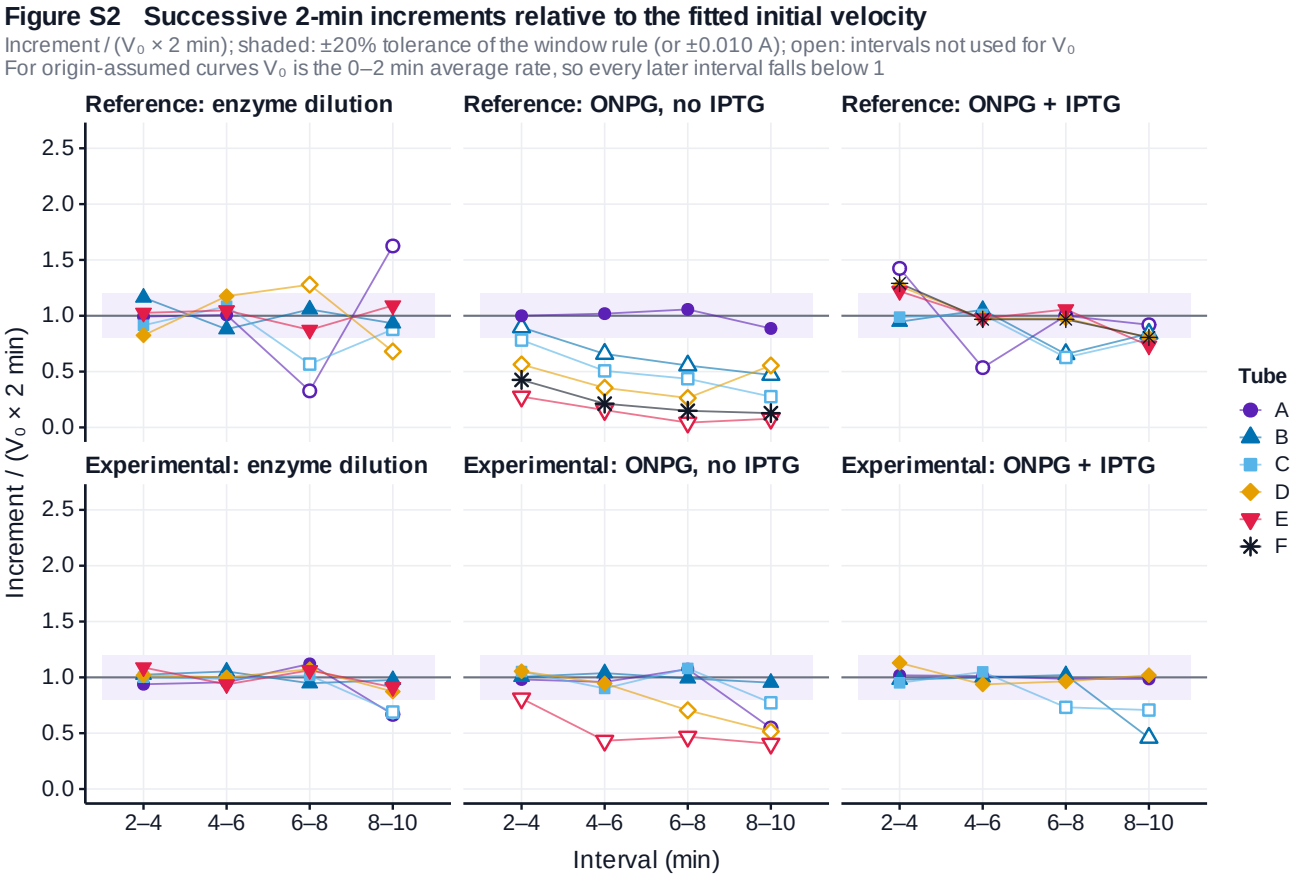
Reaction scheme, assay logic and the two saturation curves of the reference dataset with the headline values; the competitive model's shared V_{max} is a model assumption. The K_i values from the alternative initial-velocity estimators are method-dependent estimates, not a confidence interval.

Figure S1. Progress curves of the two datasets on common axes



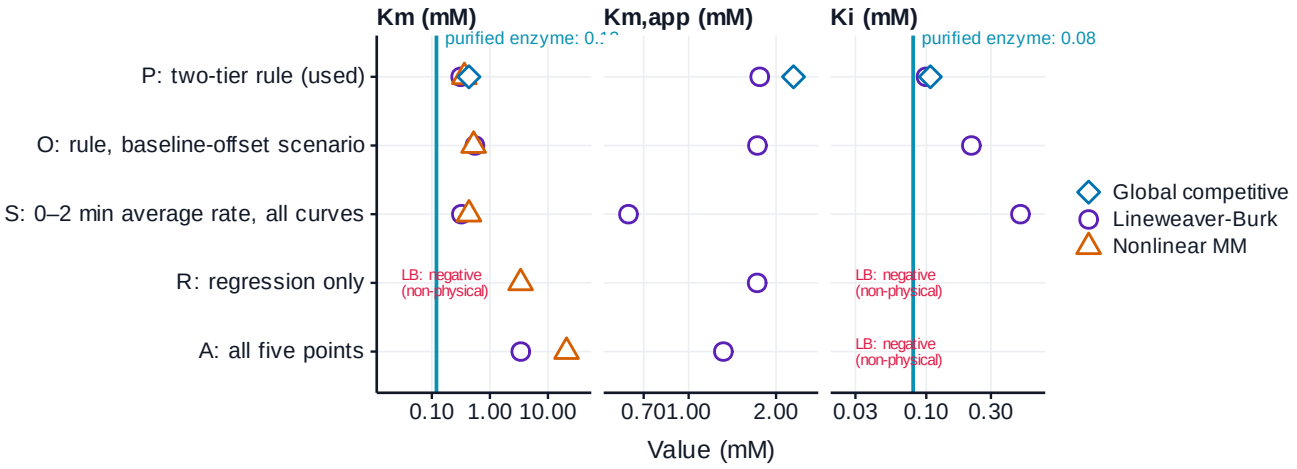
Reference dataset (left) and experimental dataset (right) for the three experiments; solid segments, regression windows; dashed, origin-assumed 0–2 min average rates; open, not used for V_0. The experimental inhibited tube B (open) is excluded from fits; inhibited tube E has no data; tube F was not performed in the experimental sessions.

Figure S2. Increment diagnostics behind every V_0 decision



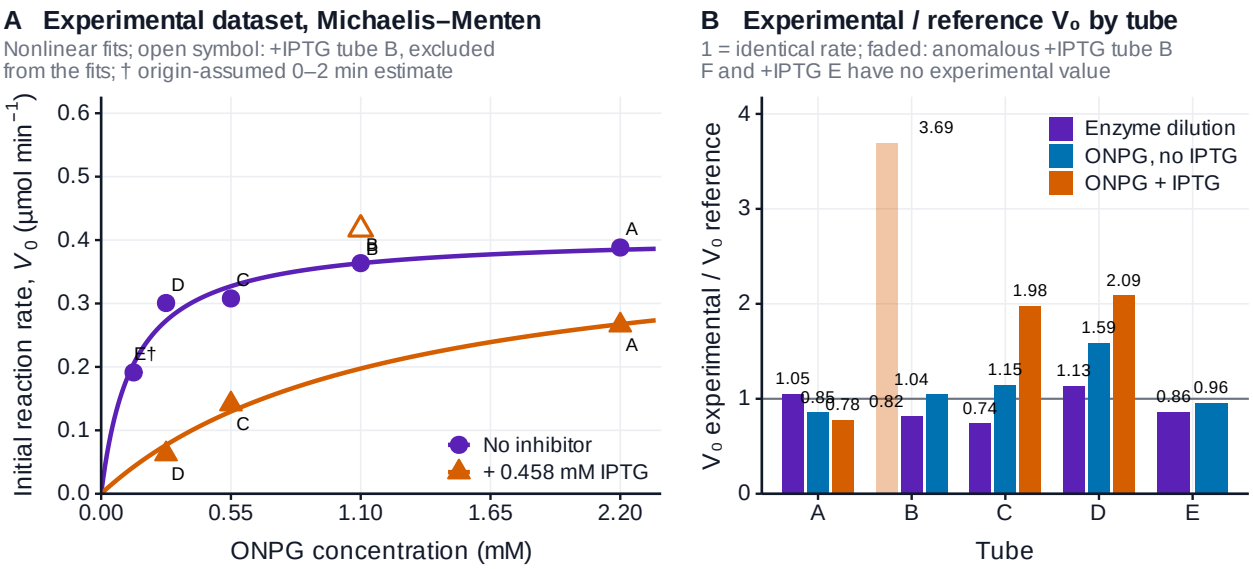
Each 2-min increment divided by the increment predicted by the fitted V_0; shaded band, acceptance window of the rule (±20% or ±0.010 A); open, intervals not used. For origin-assumed curves the reference rate is the 0–2 min average rate, so later intervals fall below 1.

Figure S3. Sensitivity of the kinetic parameters to the initial-velocity estimator



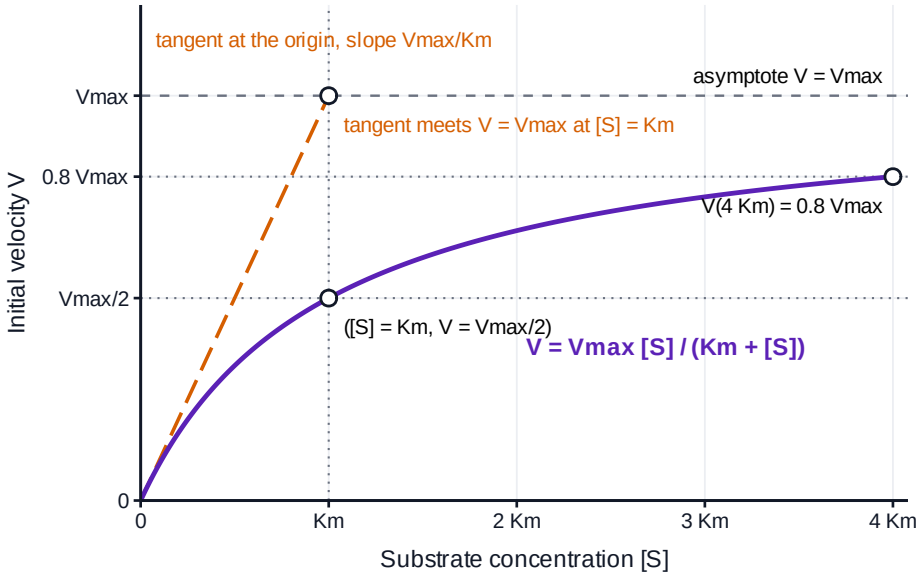
K_m, $K_{m,app}$ and K_i of the reference dataset under five estimators (A, all five points; R, regression only; S, origin-assumed 0–2 min average rate for every curve; O, rule under the baseline-offset scenario; P, the rule used); the spread is method-dependent, not a confidence interval. Cyan lines, values for purified *E. coli* β-galactosidase (TES buffer, pH 7.0, 25 °C).

Figure S4. Kinetic analysis of the experimental dataset



(A) Michaelis–Menten representation of our Part III data (nonlinear fits; open, excluded inhibited tube B). (B) Ratio of our V_0 to the reference V_0 for every tube measured in both datasets.

Diagram for Question 1 (schematic, $K_m = V_{max} = 1$)



Michaelis–Menten curve to [S] = 4 K_m with the asymptote V = V_{max}, the line [S] = K_m and the tangent at the origin (slope V_{max}/K_m); V(4K_m) = 0.8 V_{max}. This computer-drawn diagram is a reference, not the hand-drawn sketch the course requires.